

Robotic Haircutting Systems: A Survey of Methods, Challenges, and Hair Modeling Insights

Ameer Tamoor Khan  and Shuai Li , *Fellow, IEEE*

Abstract—The field of robotic haircutting is rapidly evolving, merging advances in service robotics, computer vision, force sensing, and deep learning. This survey article highlights the growing importance of robotic systems in personal grooming, driven by demographic shifts, technological innovation, and increasing demand for hygienic, consistent services. We review the historical evolution of robotic haircutting technologies, including DIY solutions, academic prototypes, and emerging commercial systems. Core technologies, such as robotic arms, hair modeling, motion planning, and tactile sensing are discussed in detail. Key challenges including hair variability, real-time decision-making, user customization, and safety considerations are analyzed. Furthermore, we explore the transformative role of virtual hairstyle modeling and the integration of multimodal perception and AI-driven personalization. Future research directions emphasize the incorporation of large language models, vision-language models, reinforcement learning, and diffusion models to advance human–robot collaboration in salons, while addressing ethical, societal, and inclusivity considerations.

Index Terms—Computer vision, deep learning, force sensing, human-robot collaboration, large language models, robotic haircutting, virtual hairstyle modeling.

I. INTRODUCTION

THE integration of robotics into personal care services has experienced significant advancements over the past decade, with robotic haircutting emerging as a particularly promising and necessary field. The growing demand for automation in grooming tasks is driven by several key factors: aging global populations, labor shortages in service industries, and heightened awareness of hygiene and the need for contactless solutions, particularly during global health crises like the COVID-19 pandemic [1]. Robotic haircutting systems offer

several advantages, including enhanced precision, repeatability, efficiency, and consistency, while simultaneously reducing the physical demands placed on human stylists.

Early developments in robotic hairdressing included a range of DIY solutions, particularly during quarantine periods, where individuals engineered simple robotic systems to assist with basic haircutting tasks [2]. At the same time, academic and industrial research initiatives proposed more sophisticated robotic platforms, integrating motion planning, tactile sensing, and computer vision to safely and effectively manipulate human hair [3]. These efforts laid the groundwork for more intelligent and autonomous haircutting systems.

Recent breakthroughs in hair modeling technologies have significantly advanced the ability of robots to interact precisely with the complex geometry and dynamic nature of human hair. Notable contributions include HairNet [4], which introduced a deep learning method to reconstruct full 3-D hair geometry from a single image, improving a robot's ability to perceive and model hair structure accurately. Building upon this, Gaussian Haircut [1] proposed a dual representation technique that combines strand-based modeling with 3-D Gaussians, allowing for high-fidelity strand reconstructions from multiview data, thereby facilitating more accurate robotic planning and manipulation.

The development of generative models has further enhanced virtual hairstyle design. Haar-like features (HAAR) [2] introduced a text-conditioned generative model for 3-D strand-based hairstyles, enabling the generation of personalized haircut plans tailored to individual preferences. Similarly, HairFIT [5] proposed a two-stage framework for pose-invariant hairstyle transfer, employing optical flow-based hair alignment and a semantic-region-aware inpainting Mask to address occlusions and pose differences between source and target images. These capabilities are essential for realistic previsualization in robotic haircutting systems.

Moreover, style-based generative adversarial network (StyleGAN) Salon [6] advanced the field by presenting a multiview latent optimization framework based on StyleGAN2 and efficient geometry-aware 3D generative adversarial network (EG3D) [7] architectures, effectively addressing issues of hallucination and detail preservation across multiple viewpoints. Such innovations enable more reliable integration of virtual hairstyle planning into robotic haircutting pipelines, offering users a preview of the final result before execution.

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Ameer Tamoor Khan is with the Department of Plant and Environmental Sciences, University of Copenhagen, DK-2300 Copenhagen, Denmark.

Shuai Li is with the Faculty of Information Technology and Electrical Engineering, University of Oulu, FI-90014 Oulu, Finland (e-mail: shuai.li@oulu.fi).

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Despite these significant technological strides, robotic hair-cutting still faces considerable challenges. Key issues include adapting to the vast diversity of human hair types and styles, ensuring user safety and comfort during manipulation, managing the dynamic behavior of hair strands, and delivering aesthetically pleasing and highly personalized haircut results. Addressing these challenges requires further research into human-robot collaboration, real-time multisensory feedback, personalized hair modeling, and the seamless integration of generative hairstyle priors.

In addition, considerations around user experience, system affordability, regulatory standards, and ethical data handling will play vital roles in the successful deployment of robotic hair-cutting solutions. As the field continues to evolve, interdisciplinary collaborations between robotics [8], computer vision [9], artificial intelligence [10], human-centered design, and cosmetology will be essential.

This survey aims to provide a comprehensive overview of the current state of robotic hair-cutting technologies from both robotic manipulation and hair modeling perspectives, identify critical gaps, and highlight future research directions that are essential for advancing the field toward widespread real-world adoption.

The rest of this article is organized as follows. Section II provides background and reviews related work in service robotics, grooming tasks, and virtual hairstyle modeling. Section III discusses innovative approaches to robotic hair-cutting, including wearable devices, robotic arms, and teleoperated systems. Section IV presents the core enabling technologies, such as robotic manipulation, motion planning, and sensory feedback integration. Section V reviews existing robotic hair-cutting systems, highlighting DIY, academic, and commercial efforts. Section VI identifies and analyzes the key challenges faced by current systems. Section VII explores the role of virtual hairstyle remodeling and planning technologies. Section VIII outlines promising future research directions. Section IX offers a critical discussion of the field's status and future needs. Finally, Section X concludes this article with a summary of insights and opportunities.

II. BACKGROUND AND RELATED WORK

The development of service robots within personal care domains has undergone remarkable transformation over the past few decades. Initially designed for basic household tasks, robotic systems are now progressively entering healthcare, elderly assistance, and grooming services. Among these, robotic hair-cutting has emerged as a promising frontier, aiming to deliver accessible, hygienic, and consistent grooming experiences.

Motivated by the necessity for contactless solutions during the COVID-19 pandemic, early efforts in robotic hair-cutting featured a variety of DIY prototypes [1]. These pioneering endeavors, although rudimentary, underscored critical challenges, such as real-time control, adaptability to diverse hair types, precision, and above all, user safety.

A key aspect of ensuring safety in human-robot interaction, particularly in grooming tasks, is force modeling. The characterization of contact forces involved in hair brushing, face wiping, and shaving activities has been systematically studied,

providing essential safety thresholds that inform the design of robotic grooming systems [11].

In academic research, more sophisticated prototypes have pushed the field forward. The University of Tokyo's Front Hair Styling Robot [3] introduced advanced capabilities, such as path planning and root-centric strand manipulation, representing a significant leap toward autonomous hairstyling. Concurrently, MIT CSAIL's RoboWig project [12] demonstrated safe and effective hair brushing through the use of sensorized brushes, emphasizing gentle contact and adaptive force regulation.

While hair modeling itself is not the central focus of robotic hair-cutting, advances in this domain provide essential priors for perception and planning modules. HairNet [4] proposed a convolutional neural network capable of reconstructing 3-D hair geometry from a single 2-D image, offering critical perceptual capabilities that enhance a robot's understanding of hair structure. HairCLIP [13] introduced a user-friendly framework for intuitive, text- and image-based hairstyle editing, facilitating the design of customized haircut plans.

Efficiency and real-time operation are crucial for practical robotic grooming. Edge- and histogram-guided generative adversarial network (EHGAN) [14] advanced real-time hairstyle transfer techniques, making them suitable for integration into grooming pipelines. GroomGen [15] contributed a hierarchical latent space representation for realistic 3-D hair strand generation, enabling more authentic simulations crucial for robotic decision-making.

Recent methodologies further enrich these prior models. Hierarchical semantic consistency network (HSC-Net), which leverages a combination of mask region-based convolutional neural network (R-CNN) and generative adversarial network (GAN) architectures augmented by a pyramidal attention mechanism, has been proposed to achieve highly accurate hairstyle transfer and segmentation [16]. A volumetric variational autoencoder capable of synthesizing 3-D hair models from single images without extensive hairstyle databases has also been introduced to improve the robustness of hair reconstruction under varied real-world conditions [17]. Neural Haircut [18] presented a two-stage strand-based reconstruction approach incorporating a diffusion prior, enabling high-fidelity hair modeling from monocular video even under uncontrolled lighting conditions.

Collectively, these advancements in hair modeling, though originally intended for virtual rendering, are increasingly essential as prior knowledge for robust, adaptive, and safe robotic hair-cutting operations. Ongoing research efforts continue to bridge gaps between perception, planning, and human-centered design, moving the field closer to practical, real-world deployment.

The progression of robotic hair-cutting technologies is depicted in Fig. 1, illustrating a clear transition from early DIY robotics experiments toward more refined academic prototypes and emerging commercial systems. This evolutionary path highlights the steady improvements in mechanical design, perception, planning algorithms, and safety mechanisms, each contributing to the growing maturity and reliability of robotic hairdressing solutions.

III. APPROACHES TO AUTOMATED HAIRCUTTING

This section explores various innovative strategies that have been proposed and implemented to advance robotic hair-cutting

Evolution of Robotic Haircutting

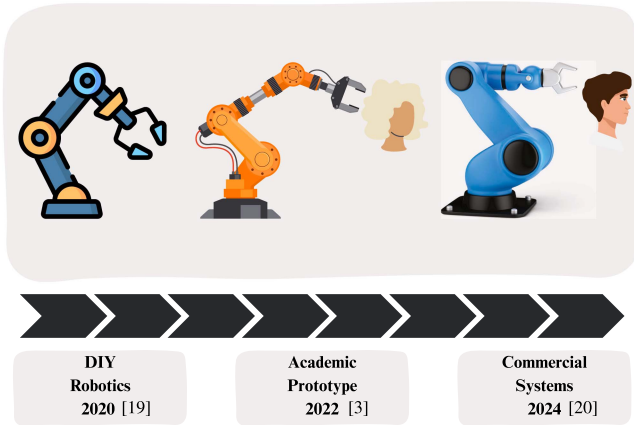


Fig. 1. Evolution timeline of robotic haircutting systems: Starting from DIY robotics experiments (2020) [19], progressing to academic prototypes (2022) [3], and moving toward emerging commercial systems (2024) [20]. Each stage marks increased sophistication in sensor integration, AI planning, and system reliability.

technologies. We discuss a range of solutions, from imaginative wearable devices and robotic arms to teleoperated systems and experimental aerial platforms, highlighting their unique contributions and challenges.

A. “Shaving Helmet” Concept

One of the most imaginative early concepts in automated grooming is the “Shaving Helmet,” developed by Thinkmodo for a viral campaign promoting HeadBlade razors [21]. Though ultimately revealed as a hoax, the concept sparked significant discussion around wearable grooming technology. The helmet was envisioned as a sleek, dome-shaped apparatus capable of delivering a full shave within 20 s. Internally, multiple razors mounted on adjustable tracks were imagined to conform to the user’s scalp, ensuring even and precise shaving. Additional features, such as a shaving cream dispenser and motorized, sensor-guided razors, were proposed to replicate the precision and touch of a skilled barber.

Despite its fictional origins, the “Shaving Helmet” highlighted key technological directions for wearable grooming devices, such as the integration of LiDAR, 3-D vision for scalp mapping, pressure-sensitive safety systems, and real-time haptic feedback. These ideas continue to inspire the development of future modular, wearable haircutting systems Fig. 2(a).

B. Robot Arms for Haircutting

1) *Shane Wighton’s DIY Haircutting Robots*: Engineer Shane Wighton pioneered two notable versions of DIY robotic haircutting systems. His first prototype, featured in “I Built a Robot to Cut My Hair” [22], utilized a motorized robotic arm executing preprogrammed movements to uniformly trim hair. This basic system required extensive manual calibration and lacked real-time adaptability.

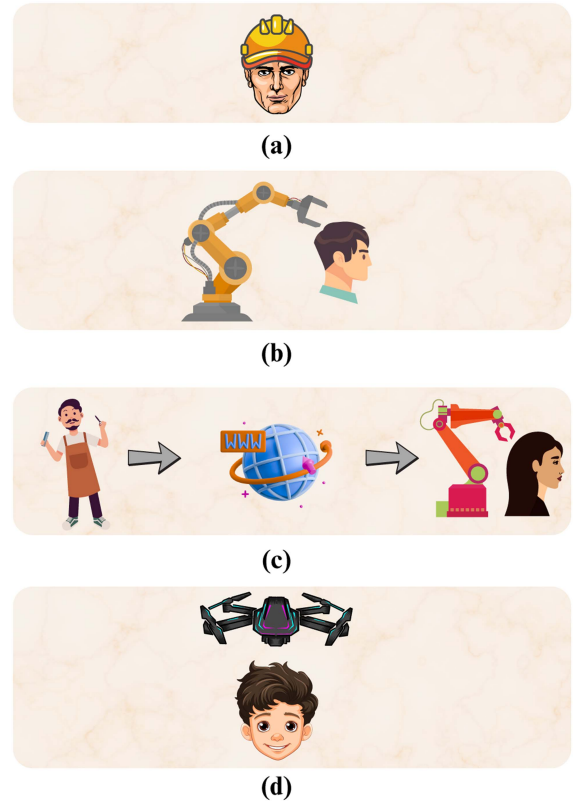


Fig. 2. Illustration of different robotic haircutting approaches. (a) Wearable “Shaving Helmet” concept for automated grooming. (b) Robotic arms performing haircutting based on scanning and planning. (c) Tele-operated haircutting involving a remote human expert controlling a robot. (d) Experimental drone-based haircutting demonstrating future possibilities.

Wighton’s second iteration [23], shown in “I Upgraded My Haircut Robot,” addressed previous limitations by incorporating computer vision and real-time head tracking. By scanning the user’s scalp shape and adjusting the cutting trajectory dynamically, this version offered smoother, more precise cuts adaptable to different hair lengths and densities Fig. 2(b).

2) *3-D Scanning-Based Robotic Haircutting*: A more advanced method involves a fully autonomous robotic arm guided by a smartphone-based 3-D scanning system [24]. By capturing a detailed 3-D model of the user’s head, the robot can plan a customized haircut path tailored to individual contours. Equipped with collision detection and controlled cutting pressure, these systems mark significant strides toward safe, precise, and customized robotic haircuts, showing how consumer-grade technology can enhance robotic grooming capabilities.

3) *Collaborative Robot Haircutting*: The Sawyer collaborative robot from Rethink Robotics [25] exemplifies the use of industrial-grade cobots for personal grooming. Outfitted with an automatic trimmer and force-sensing capabilities, Sawyer can detect physical contact and adjust its path to safely and precisely maneuver around the user’s head. With seven degrees of freedom and integrated vision systems, Sawyer demonstrates how cobots can bridge human dexterity and robotic efficiency in grooming applications.

C. Teleoperated Haircutting Systems

1) *NBC News Demonstration*: A notable demonstration featured a skilled barber remotely operating an industrial robotic arm to perform a live head shave [26]. Cameras and sensors provided the barber with real-time feedback on head contours, enabling precise and safe shaving. Though not autonomous, this teleoperated system combined human expertise with robotic precision, showcasing a potential hybrid approach to remote personal grooming services Fig. 2(c).

2) *High-Precision Teleoperation*: In another demonstration [27], a user teleoperated a robotic arm equipped with scissors to cut their own hair with remarkable precision. Despite operating without direct line-of-sight, the user achieved a professional-quality haircut, underscoring the viability of teleoperation for personal grooming tasks where fine motor control is essential.

D. Inspirations From Panasonic's Shampoo and Head Massage Robot

Although not designed for haircutting, Panasonic's shampoo and scalp massage robot [28], [29] offers valuable design insights. Using articulated robotic arms with 16 "finger-like" extensions, the robot adapts its pressure and movements based on a scanned model of the user's head. Its emphasis on comfort, adaptability, and precision mirrors the goals of future haircutting robots and highlights the importance of personalized, gentle robotic interaction in personal care.

E. Exploratory Drone-Based Haircutting

Experimental attempts at using drones equipped with cutting tools [30] illustrate a futuristic, though currently impractical, approach to robotic grooming. While current drones lack the millimeter-level precision required for safe haircuts, future advancements in stabilization, sensor integration, and AI-driven path planning could potentially make drone-based haircutting feasible Fig. 2(d). Drones offer inherent flexibility and accessibility advantages, providing an exciting direction for future research in mobile grooming technologies.

IV. TECHNOLOGIES ENABLING ROBOTIC HAIRCUTTING

This section explores various innovative strategies that have been proposed and implemented to advance robotic haircutting technologies. We discuss a range of solutions, from imaginative wearable devices and robotic arms to teleoperated systems and experimental aerial platforms, highlighting their unique contributions and challenges.

A. DIY Robotic Haircutting by Stuff Made Here

A notable DIY effort was demonstrated by the YouTube creator Stuff Made Here [19], where a highly customized robotic haircutting system was built. The system utilized a UR5e robotic arm equipped with an electric hair clipper attached to the end-effector. A 3-D scanning system mapped the user's head shape, and custom path planning algorithms generated cutting paths based on the desired hair length distribution. The robot

operated semiautonomously under human supervision to ensure safety. Techniques, such as real-time head tracking, dynamic path replanning, and force-controlled contact with the scalp were implemented to maintain haircut precision while minimizing the risk of injury. This project showcased both the promise and technical challenges associated with at-home robotic haircutting systems, highlighting the need for robustness, adaptability, and user trust.

B. Robotic Arms and Actuators

Robotic hairdressers require highly precise and dexterous robotic arms capable of handling delicate tasks, such as hair manipulation, sectioning, and cutting. These robotic manipulators integrate compliance mechanisms to adapt to the flexible and dynamic nature of human hair and scalp. For example, RoboWig utilizes a compliant robotic arm outfitted with a sensorized brush to perform hair brushing safely [11]. The MOE-Hair system [31] further advances this area by employing tendon-driven soft robotic end-effectors specifically designed for haircare tasks, emphasizing mechanical compliance, safety, and user comfort. Recent advancements in actuator technology include integrated force control, soft robotics, and bio-inspired designs, ensuring safe and adaptive contact with human tissues.

C. Computer Vision and Hair Detection

Accurate perception of hair and scalp structure is critical for successful robotic haircutting. 2-D and 3-D hair modeling methods, such as HairNet [4], enable the reconstruction of full 3-D hair geometry from a single RGB image, greatly enhancing a robot's spatial understanding of hair volume, density, and orientation. Segmentation techniques based on deep learning frameworks like mask R-CNN [16] allow precise separation of hair from facial features and background, facilitating targeted manipulation. In addition, pose estimation and orientation field prediction techniques [18] enable robust hair modeling under varying head poses and lighting conditions, making the robotic system more resilient to real-world challenges.

D. Motion Planning and Control

Motion planning in robotic haircutting involves designing efficient, safe, and human-centered trajectories that manipulate hair strands without causing discomfort. Techniques, such as root-centric strand adjustment, proposed by the University of Tokyo system [3], maintain hair root stability while enabling precise styling and cutting. The design and evaluation of robotic hair combing systems [32] contribute further insights into the importance of compliant motions and adaptive grooming strategies. Effective motion planning must account for the dynamic behavior of hair, variations in hair texture and thickness, and the need for consistent user experience across different individuals.

E. Force and Tactile Sensing

Force sensing is essential to ensure safe and effective human-robot interaction during grooming tasks. Accurate modeling of allowable contact forces [11] informs the development of control

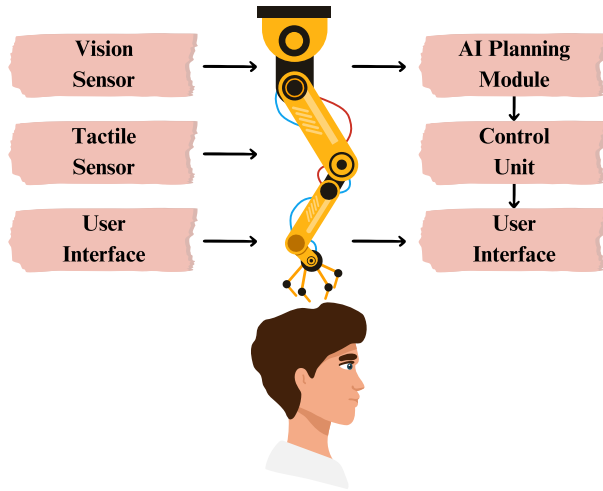


Fig. 3. System architecture of a robotic haircutting setup integrating vision sensors, tactile sensors, and user interface inputs. Sensor data are processed by an AI planning module and control unit to adaptively guide the robotic arm in delivering a personalized and safe haircut.

strategies that prevent discomfort or injury. Tactile sensors embedded into robotic end-effectors provide real-time feedback on contact pressure and hair tension, enabling dynamic adjustment of actions to maintain safe interaction. Furthermore, inclusive design principles for assistive haircare robots [33] stress the importance of ethical considerations and cultural sensitivity when designing force application strategies, ensuring that robots cater to diverse user needs.

Sensor fusion strategies are critical for reliable robotic haircutting operations. Effective multimodal perception combines visual, tactile, and force feedback to create comprehensive situational awareness. For instance, vision sensors provide geometric hair structure while tactile sensors detect physical contact and hair tension during cutting operations [31]. Force sensors complement this information by monitoring applied pressure and detecting potential entanglement scenarios. The integration of these modalities requires sophisticated algorithms that can process and fuse heterogeneous sensor data in real time, enabling responsive control and adaptive behavior [34].

F. Working Architecture Overview

Fig. 3 illustrates the overall architecture of an advanced robotic haircutting system. The system integrates several interdependent modules: a compliant robotic arm equipped with visual and tactile sensors performs hair manipulation tasks with precision. These sensors supply real-time feedback to a central control unit, responsible for path planning, force adjustment, and real-time error recovery. In parallel, an AI-based planning module receives user input through a graphical interface and dynamically generates personalized haircut plans based on individual hair properties and styling preferences. Continuous feedback loops between sensing, planning, and actuation ensure adaptive, safe, and user-specific haircutting processes. This modular and hierarchical design architecture significantly enhances safety,

personalization, scalability, and overall robustness of robotic hairdressing solutions.

V. EXISTING ROBOTIC HAIRCUTTING SYSTEMS

In this section, we review the current landscape of robotic haircutting systems, spanning from early DIY innovations to academic research prototypes and emerging commercial solutions. These developments collectively demonstrate the technological progress, creativity, and challenges faced in automating hair grooming tasks.

A. DIY Robotic Haircutting Solutions

During the COVID-19 pandemic, DIY robotic haircutting solutions emerged as innovative responses to the lack of access to salons. Individuals built simple robotic systems using robotic arms, 3-D-printed parts, and electric clippers, controlled via custom software or remote teleoperation [27]. Although rudimentary, these systems demonstrated the feasibility of automating haircutting and identified key challenges, such as safety, adaptability, and hair detection. These projects served as important proofs-of-concept, inspiring future research directions in this field.

B. Academic Prototypes

Academic research has developed more sophisticated robotic haircutting systems aimed at improving precision, safety, and personalization. The University of Tokyo's Front Hair Styling Robot System [3] introduced advanced path planning techniques focused on root-centric strand manipulation, enabling controlled hair adjustments toward target styles.

Another notable example is RoboWig from MIT CSAIL [12], which focuses on hair brushing tasks as a precursor to more advanced grooming. RoboWig employs a compliant robotic arm equipped with a sensorized brush, leveraging both visual and force feedback to safely brush and detangle hair. This work lays critical groundwork for the development of future haircutting and hairstyling robots.

C. Commercial and Experimental Salon Robots

While fully autonomous robotic haircutting remains under development, commercial and experimental efforts are beginning to surface. Some experimental salon robots equipped with scissors or razors have been showcased at technology expos [20]. These systems typically operate in semiautonomous modes, combining robotic actuation with human oversight to ensure safety and accommodate individual customization. Current commercial systems include prototype installations in select salons in Japan and South Korea, where robotic arms assist with basic trimming operations under strict supervision. However, market adoption remains limited due to regulatory concerns and customer acceptance challenges [35]. Companies are increasingly exploring integrations of AI-based hairstyle planning and robotic manipulation to deliver more personalized services, as summarized in Table I.

TABLE I

SUMMARY OF EXISTING ROBOTIC HAIRCUTTING SYSTEMS: CATEGORIZATION OF DIY, ACADEMIC, AND COMMERCIAL ROBOTIC HAIRCUTTING SOLUTIONS, HIGHLIGHTING THEIR KEY TECHNOLOGIES AND LEVELS OF AUTONOMY

System	Type	Key Technologies	Automation Level
DIY Barber Robot [1]	DIY	Basic motor control	Manual/Semiautonomous
University of Tokyo Robot [3]	Academic	Root-centric path planning	Semiautonomous
RoboWig (MIT CSAIL) [12]	Academic	Visual + tactile sensing	Semiautonomous
Salon Robots [20]	Commercial	Scissors/razors automation	Human-supervised

VI. KEY CHALLENGES

This section outlines the major challenges hindering the widespread adoption of robotic haircutting systems. Despite impressive technological progress, several fundamental issues must be addressed to ensure safe, effective, and user-friendly solutions.

First, the variability in human hair characteristics presents a substantial obstacle. Hair differs widely among individuals in terms of texture (e.g., straight, wavy, curly, coiled), length, thickness, density, and moisture content. These differences dramatically influence how hair behaves under mechanical manipulation. For example, curly hair tends to entangle easily, requiring more nuanced strategies compared to straight hair. In addition, environmental factors, such as humidity can dynamically alter hair properties, necessitating highly adaptive robotic systems [11].

Second, real-time sensory feedback and decision-making are critical for safe and effective haircutting. Robots must quickly interpret multimodal sensory inputs—including force, vision, and tactile data—to respond to unpredictable user movements, sudden shifts in hair behavior, and unforeseen events (e.g., tool entanglement) [31]. Many existing systems still lack the ultra-fast feedback loops needed for seamless and safe adaptation.

Third, ensuring safety, comfort, and intuitive human–robot interaction is essential. Robots must operate within safe force thresholds, maintain smooth and gentle contact with the scalp, and avoid accidental injuries [33]. Effective communication through real-time feedback and visual or auditory cues is also necessary to promote user trust and engagement.

Fourth, system failures and unexpected scenarios pose significant challenges. Current robotic haircutting systems demonstrate limited capability in handling sudden user head movements, which can lead to uneven cuts or safety hazards. Hair entanglement scenarios, where strands become twisted around cutting tools or robotic joints, often require immediate human intervention. In addition, the tradeoffs between cutting speed and safety remain poorly optimized, with faster operations typically compromising precision and user comfort [36].

Finally, user customization and aesthetic preferences must be at the forefront of system design. Robotic haircutting systems need to accommodate a wide variety of cultural and personal hairstyle choices, face shapes, and evolving fashion trends. Developing flexible customization pipelines that translate user intent into precise robotic actions remains an open research challenge [5].

Ethical considerations around data privacy present additional challenges. User scalp scans and hairstyle preferences constitute sensitive personal data that requires secure handling and storage. Current systems lack comprehensive frameworks for addressing algorithmic bias in hairstyle recommendations, particularly concerning diverse hair types across different ethnic and cultural backgrounds. Studies examining how robotic systems perform across various hair textures and cultural styling practices are notably absent from current literature [37], [38].

VII. HAIRSTYLE REMODELING AND PLANNING

In this section, we examine the role of virtual hairstyle remodeling and planning technologies in the robotic haircutting pipeline. These tools bridge the gap between user preference and robotic execution, providing a means for users to visualize, customize, and approve hairstyles prior to physical implementation.

Virtual hairstyle remodeling and planning have emerged as crucial components in modern robotic haircutting workflows. These technologies allow users to preview potential hairstyles before committing to physical cuts, significantly improving user satisfaction.

HairFIT [5] provides a framework for pose-invariant hairstyle transfer using flow-based hair alignment and semantic-region-aware inpainting. It addresses common challenges, such as occlusions and pose differences between source and target images. Users can visualize different haircuts superimposed on their own images, making informed style choices.

Similarly, StyleGAN Salon [6] introduces a multiview latent optimization technique built upon StyleGAN2 and EG3D architectures. By considering multiple viewing angles and optimizing latent representations, StyleGAN Salon produces highly realistic and consistent hairstyle transformations across diverse poses and lighting conditions.

Integration of these virtual techniques with robotic hardware enables a seamless transition from digital visualization to physical execution [20]. After selecting a desired hairstyle virtually, the robotic system can plan the necessary cutting and styling trajectories based on the selected design. This approach not only enhances the personalization of robotic haircutting but also reduces the anxiety and uncertainty typically associated with salon visits.

Continued research is focusing on improving the realism of virtual projections, handling rare or complex hairstyles, and automating the translation of digital models into actionable robotic paths, as summarized in Table III.

TABLE II

KEY CHALLENGES IN ROBOTIC HAIRCUTTING: IDENTIFICATION OF MAJOR TECHNICAL AND USER-CENTERED CHALLENGES IMPEDING THE WIDESPREAD DEPLOYMENT OF ROBOTIC HAIRCUTTING SYSTEMS

Challenge	Description
Hair Variability [11]	Different textures, densities, and hair behaviors across individuals and environmental conditions require adaptive robotic manipulation strategies.
Real-time Feedback [31]	The robot must integrate sensory inputs rapidly and make dynamic decisions for responsive control during haircutting operations.
Safety and Comfort [33]	Maintaining safe force levels and providing comfortable interactions are essential for user trust and injury prevention.
User Customization [33]	Systems must support a wide range of user styling preferences, cultural hair practices, and aesthetic requirements.
System Failures and Limitations [36]	Handling unexpected scenarios such as sudden user movements, hair entanglement, and tool failures requires robust error recovery mechanisms.

TABLE III

TECHNOLOGIES FOR VIRTUAL HAIRSTYLE REMODELING: OVERVIEW OF RECENT VIRTUAL HAIRSTYLE TRANSFER AND EDITING TECHNOLOGIES, DETAILING THEIR STRENGTHS AND LIMITATIONS FOR INTEGRATION INTO ROBOTIC HAIRCUTTING WORKFLOWS

Method	Input Type	Strengths	Limitations
HairFIT [5]	Image pairs	Provides pose-invariant hairstyle transfer with natural occlusion handling via semantic inpainting.	May struggle with highly complex or rare hairstyles.
StyleGAN Salon [6]	Single/multiview images	High-fidelity, multiview latent optimization enabling robust hairstyle transformations across different poses.	Requires significant computational resources for optimization.
HairCLIP [20]	Text and image	Multimodal editing capabilities allow intuitive user-driven hairstyle design based on natural language and images.	Limited true 3D realism in generated outputs.

VIII. FUTURE RESEARCH DIRECTIONS

This section outlines emerging research trends that are poised to drive the next generation of robotic haircutting systems. These innovations aim to enhance perception, learning, personalization, and human–robot collaboration.

The future of robotic haircutting lies in the convergence of multiple cutting-edge research areas that leverage deep learning, multimodal perception, and advanced user personalization techniques.

First, multimodal perception [31] will be vital. Combining visual, tactile, auditory, and potentially olfactory data could dramatically improve a robot’s ability to perceive hair properties, user emotional states, and environmental factors. For instance, combining vision and tactile sensors may allow robots to better judge hair tangling, adjust brushing force dynamically, and improve trajectory planning in real-time [34].

Second, learning from demonstration (LfD) and reinforcement learning (RL) [39] approaches can revolutionize how robots acquire complex grooming skills. Robots could learn haircutting strategies directly from observing expert human stylists, incorporating nuances like wrist movement, scissor handling, and tool angle. RL in simulated environments can enable robots to refine these strategies iteratively without risking human injury [40].

Modern diffusion models, such as Stable Diffusion [41] and GANs [42] offer new opportunities for realistic hair modeling and style generation. These models can be used not only to enhance virtual hairstyle rendering but also to train robots on synthetic datasets of hair manipulation scenarios, dramatically reducing the need for costly real-world data collection [43].

Large language models (LLMs) like GPT-4 [44] and DeepSeek [45] can enhance human–robot interaction by providing natural language interfaces. A potential framework for LLM integration involves a multistage translation process: 1) natural language parsing to extract styling intent (e.g., “trim two centimeters from the top and style into a fade”); 2) semantic mapping to geometric parameters and tool selections; and 3) trajectory generation for robotic execution. For example, the phrase “fade cut” would be interpreted as gradual hair length transitions, mapped to specific clipper guard sequences, and translated into path planning algorithms that control cutting depth and movement patterns [46].

Vision-language models (VLMs), such as Flamingo [47] and bootstrapping language-image pre-training, version 2 (BLIP-2) [48], combine textual and visual understanding, enabling robots to better interpret customer-provided images (e.g., “style like this photo”) and adjust plans accordingly. VLMs can support advanced consultation systems that predict user preferences and suggest hairstyle options based on uploaded images and text input.

Third, developing personalized haircut models [15] based on individual user data (e.g., hair type, scalp sensitivity, preferred styles, historical haircuts) will enable systems to offer highly customized grooming services. Deep personalization will ensure that robotic services adapt to each individual’s unique characteristics over time.

Finally, fostering collaboration between robots and human stylists [33] presents an exciting vision for future salons. Robots can handle repetitive tasks like precise trimming and sectioning,

TABLE IV

EMERGING DEEP LEARNING TOOLS FOR ROBOTIC HAIRCUTTING: SURVEY OF NEXT-GENERATION AI MODELS AND FRAMEWORKS THAT COULD ENHANCE PERCEPTION, PERSONALIZATION, AND CONSULTATION IN ROBOTIC HAIRCUTTING

Tool	Application Area	Notable Features
Stable Diffusion [41]	Hair simulation, dataset generation	Capable of producing high-quality synthetic images for training or virtual hairstyle projection; supports latent-based fine-tuning.
GPT-4 [44]	Natural language consultation	Enables complex dialogue systems for robotic haircut consultation and instruction following.
BLIP-2 [48]	Vision-language interpretation	Integrates vision and text, enabling robots to process hairstyle references provided as images and descriptive prompts.
Flamingo [47]	Few-shot multimodal learning	Allows rapid adaptation to new tasks with minimal examples, useful for quickly learning new hairstyle types or grooming behaviors.
DeepSeek [45]	Open-source language modeling	Provides scalable and customizable LLMs that can be fine-tuned for haircut-specific consultation or planning tasks.

allowing human stylists to focus on creative, high-level styling decisions [49]. Such hybrid systems could significantly improve productivity, quality, and customer satisfaction.

Future work must also address regulatory, ethical, and societal impacts, including privacy preservation when collecting user hairstyle data, algorithmic fairness across diverse hair types, and ensuring accessibility across global cultures, as summarized in Table IV.

IX. DISCUSSION

The survey of robotic haircutting technologies reveals both remarkable progress and persistent challenges across mechanical design, perception, planning, and human interaction. Early DIY projects demonstrated the feasibility of automated grooming but also highlighted critical issues of adaptability, safety, and precision. Academic prototypes have significantly advanced the field by integrating compliant manipulators, force sensing, and computer vision, yet full autonomy and robustness remain distant goals.

A notable trend is the increasing reliance on multimodal sensory integration, combining vision and tactile feedback to enhance situational awareness during grooming tasks. However, achieving real-time, closed-loop control with the precision necessary for haircutting still poses significant computational and algorithmic hurdles. Furthermore, existing systems often focus on brushing and surface grooming rather than complex, high-precision cutting and styling operations, indicating a gap in current capabilities.

Virtual hairstyle remodeling and planning technologies offer a promising bridge between user intention and robotic execution, improving user trust and personalization. Nevertheless, most systems still struggle with modeling rare hairstyles and dynamic hair behaviors under varying conditions. There is also a need for more seamless integration between virtual planning platforms and physical robotic execution modules.

From an ethical and societal perspective, future robotic haircutting systems must prioritize inclusivity, ensuring equal performance across diverse hair types and cultural styles. Addressing privacy concerns related to user data collection and maintaining algorithmic fairness will be critical for widespread adoption. In addition, considerations must be given to accessibility for individuals with disabilities, the risk of reinforcing aesthetic biases through AI-driven hairstyle recommendations, and the need for transparent user consent in hairstyle data usage.

Ethical frameworks must evolve alongside technical innovation to ensure that robotic haircutting technologies serve all communities equitably and responsibly.

Overall, while foundational building blocks have been established, achieving robust, fully autonomous, and user-personalized robotic haircutting will require major advances across multimodal sensing, LfD, deep generative modeling, and human-robot collaboration.

X. CONCLUSION

In this survey, we presented a comprehensive overview of the landscape of robotic haircutting systems. Beginning with the motivation for automation in personal grooming, we explored the significant progress made through DIY experiments, academic research, and emerging commercial solutions.

We highlighted the critical core technologies that underpin robotic haircutting, including compliant robotic arms, advanced computer vision for hair modeling, motion planning strategies, and real-time tactile feedback mechanisms. The integration of virtual hairstyle remodeling tools, such as HairFIT and StyleGAN Salon has also opened new possibilities for personalized haircutting experiences.

Despite the promising advancements, key challenges remain. Addressing human hair variability, ensuring user comfort and safety, achieving real-time sensory processing, and enabling aesthetic personalization are vital for widespread adoption. Moreover, the ethical implications and cultural sensitivities associated with personal grooming robots must not be overlooked.

Looking forward, the fusion of multimodal perception, large-scale generative models like Stable Diffusion, LLMs, such as GPT-4, and vision-language frameworks like Flamingo and BLIP-2, holds enormous potential. These technologies could enable robots not only to perform haircuts but to consult, plan, and adapt dynamically to user preferences in natural, human-like ways.

Ultimately, robotic haircutting is poised to transform personal care, providing safer, more accessible, and highly personalized services. Rather than replacing human stylists, future robotic systems are likely to collaborate with them, enhancing creativity, efficiency, and client satisfaction in hybrid salon environments. Through continuous innovation and interdisciplinary research, robotic haircutting will become an integral part of the evolving landscape of personal grooming technology.

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